Azer Omar

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Education

Linköping University Linköping, Sweden
Bachelor of Science in Computer Engineering *Aug. 2022 – jun 2025*

Projects

bachelor thesis: FPGA based AES-128 | *VHDL, FPGA, Soft-Core, Timing Analysis* march. 2025 – current

- Designed and implemented AES-128 encryption on both a soft-core processor and a dedicated FPGA logic design
- Compared performance in terms of memory usage, area (resource utilization), and speed (latency and throughput)
- Used simulation tools and synthesis reports to quantify differences and evaluate trade-offs in architectural approaches
- Documented findings in a comparative study highlighting benefits of hardware acceleration over general-purpose processing

Project Course TSEA29 | *Git, C/C++, AVR, Raspberry pi, UART, SPI, PID* Sep. 2024 – Nov. 2024

- Developed a robot to work in a warehouse environment
- Setup communication between different modules using SPI and UART
- Automatic pathfinding through a given map
- Developed a PD controller to keep the robot stable on the given map

Project Course TSIU03 | *FPGA, Quartus software, VHDL* aug. 2024 – sep. 2024

- Built a oscilloscope displaying the music waves from a phone
- displayed the values read from a ADC as a wave onto a VGA screen
- added the ability to increase and decrease the volume from a keyboard input

LIU Formula Student AMS engineer | *Git, Arduino, STM32, CAN, C/C++* Oct. 2022 – Nov. 2023

- Responsible for the Battery Management System (BMS) for the LIU Formula Student electric car 2023, where Arduino was used to monitor battery performance and activate safety mechanisms.
- Developed software that meticulously tracked temperature and voltage levels, enhancing the vehicle's safety and reliability.

Technical Skills

Languages/skills: Python, C/C++, JavaScript, VHDL, Node.js, Flask, react, HTML, css
Developer Tools: Git, Github, VS Code, nvim, linux